

GIL

A Generally Intelligent Learner

A cognitive system instantiating the Piagetian Modeler —
a mind that perceives a world, models it, wants things, acts, and learns.

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A blueprint, and a thing that runs

Two words that are not interchangeable.

Piagetian Modeler — the ARCHITECTURE

A blueprint. A theory of what a mind is made of and how its parts relate.

It says a mind has a Totality, an Observation component, Coordination, Reflection, Consolidation — and how knowledge is constructed by a subject acting on a world.

GIL — the SYSTEM

The physical implementation. A thing that runs.

It holds a knowledge base, occupies ports, registers devices, spends actions, and can be pointed at a world. Other systems could instantiate the same architecture; each would be a distinct mind.

Named for Jean Piaget — whose account of intelligence this implements rather than merely cites

scheme — what is repeatable and generalizable in an action

assimilation — taking the world into what you already know

accommodation — changing what you know because the world would not fit

equilibration — the moving balance between the two — which is intelligence

Not a model. A mind.

The difference is architectural, not rhetorical.

A language model

Answers when asked.

Has no world, no body, no goals of its own.

Cannot tell you why it concluded anything — the reasoning is not a structure, it is a trajectory.

Forgets between sessions unless something outside it remembers.

A GIL mind

Perceives continuously, whether or not asked.

Holds a world model, forms urges, poses goals, acts, and grades what came back.

Every conclusion is a row — the percept, the match, the score, the scheme, the belief, the attempt.

Remembers across years. Its knowledge base is the mind.

Five components

Each a population of mechanisms, not a module with an API.

1

Totality

The knowledge base. Percepts, schemes, cases, beliefs, activations — everything the mind holds.

2

Observation

Perceiving. Detectors, Matcher, Storer, Activator. Turns what arrives into what is known.

3

Coordination

Wanting and acting. Urges, agendas, deliberation, ten solvers, the Executor.

4

Reflection

Thinking about itself. Simulation, exploration, imitation, play, monitoring.

5

Consolidation

Keeping house. Compression, automaticity, forgetting, skill formation.

Nothing calls anything

Mechanisms fire on what they find and leave results where the next one looks.



This is stigmergy — the coordination of termites, and of neurons.

No dispatcher. No pipeline. No orchestration layer to edit when a mechanism is added.

279 rules and forty-odd mechanisms run in parallel over one shared structure. Each fires when its conditions are met; what it leaves behind is what the next one reads. Several mechanisms bear on one problem conjunctively, each doing the part its conditions match.

Which is why the architecture scales: adding a mechanism does not mean editing a controller.

The monad

One identity, across perceiving, remembering, imagining and reasoning.

128

bits, signed

A monad is what the mind refers to

Not a pointer, not a row id. It is the identity of a thing known — minted once by the reifier, and the same whether the thing is being seen now, recalled from last year, imagined, or reasoned about.

Its negative is its dyad

(- monad) is the negation — a separate identity with its own belief and its own activations.

So a mind can be uncertain about a claim AND its opposite, which is honest ignorance; or confident of one, which is knowledge. A single signed number could not tell those apart.

Perception is flat. Imagination is solid.

Two parallel families, joined but never mixed.

PERCEPTUAL — 2D, seen

Figure a region in a frame

Spot where it was found

Scene one arrangement, witnessed

Clip the scenes of one visit

Venue where an activity occurs

IMAGINATIVE — 3D, staged

- **Figment** meshes and clouds

- **Ambit** where the mind put it

- **Taxis** one arrangement, staged

- **Animation** a sequence of shots

- **Canvas** the stage surface

Staging is the only mechanism that crosses — and the only one that infers

It reads a Scene of Figures and builds a Taxis of Figments: a solid proposed for a picture. Three schemes record the inference — Embodies (this solid is that region), Evokes (this sight called up that arrangement), Depicts (this canvas is my model of that place).

A mind that can look at what it imagined

Hypothesis testing without acting.

1

Stage

Propose a solid for what was seen

2

Render

Draw it from the vantage it was seen at

3

Perceive

Hand the image back to the mind's own detectors

4

Compare

Does it resolve to what was actually there?

Piaget called this the mental experiment.

The mind builds a thing it has never seen, looks at it, and finds out whether it resolves to something known. A form recognised in a dreamt canvas is a form the mind can now recognise in the world — which is hypothesis generation at no cost in actions, risk, or consequence.

Space is built by wandering

Scene → Venue → Locale → Place → Area → Region → Map

Scene

one arrangement, witnessed

Venue

where an activity occurs — about 16 scenes, a panorama

Locale

several venues, identified by a landmark that does not move

Place

several locales

Area

several places

Region

several areas

Map

a world — and a mind holds several

Chunking is discovery, not splitting. When five to nine things belong together, the grouping is made once and closed.

One belief per scheme, per viewpoint

Viewpoint 1 is the mind's own. Every other models somebody else's.

Viewpoint 1

General Awareness — what this mind believes.

The default, so every mechanism that reasons from the mind's own position gets it without asking.

Viewpoint n

What this mind believes ANOTHER entity believes about the same monad.

Modelling another mind is the deliberate act.

Diametric

This viewpoint's belief about the dyad — the negated claim.

Not disagreement. The opposite proposition, held separately.

This is how a GIL mind holds a false belief

Sal watched the box moved and the mind did not. Two Beliefs, one monad, two viewpoints, two certainties — and the mind expects Sal to act on his. No special machinery: it is the storage layout.

Less search. More recognition.

A skill activates the actions and reasons that have reliably served goals like this one.

How a skill forms

Reliable actions — not merely attempted ones — grouped by the similarity of the goal sets they serve, within one abstraction band and consistent modalities. The objectives are never chosen; they fall out as the union of what the members serve.

How a skill is used

An objective activates. One similarity probe finds the fitting skill. It disperses to its actions and reasons — so what has worked before is already lit when deliberation begins. It primes; it does not choose.

A skill that primes everything primes nothing.

Which is why a skill must DIFFERENTIATE as it grows — opening, middlegame, endgame, each with its own narrower objectives, and the objective in hand picking the branch.

That is what expertise looks like from outside. A novice has “chess”; a master has an articulated structure of it. Differentiation is not a refinement of skill — it is what keeps a skill usable as it grows.

The Psyche

A mind has no senses of its own. A Psyche is how it meets a world.



Four tuple types, one door. The architecture does not change when the world does — the same Perceiver handles a camera frame, a Slack message, a stock tick and a torque reading.

The possible worlds

A world is anything that can be perceived and acted upon.

Eidos

εἶδος

The ARC-AGI-3 grid. 64×64, 16 colours, 7 actions, no instructions.

Expanse

the territory

Physical reality. Figure, Optimus, NAO, Unitree G2 — one interface.

Aether

the medium

The internet. Services, files, processes, messages, browsers, APIs.

Kojin

個人

One human being as world — what they say, want, can do, and avoid.

Shudan

集團

One organization as world — meetings, records, metrics, departures.

Gotham

the open city

Liberty City. Persistent, populated, physically simulated, unpausing.

Same mind. Same mechanisms. Different Psyche.

Shudan — the organization as world

集団 · to gather, and the bounded body. The collective as a single subject.

Discourse	meetings, messages, retrospectives, what is avoided	Record	documents, metrics, decisions, stated policy
Structure	org changes, roles, unit state, attrition	Preference	what is stated, and what budgets actually reveal
Capability	strengths and weaknesses, with evidence	Environment	opportunities and threats, scored against both
Depth	ego traits, shadow traits, scapegoating, values gaps	System	authorization, sessions, connection status

The Shudan infers an organization's shadow from what it systematically cannot discuss — topics avoided across venues, units scapegoated for problems the whole body created, the gap between “we hire for potential” and pedigree-weighted hiring.

Opportunities scored only against the ego model produce strategies the organization cannot execute.

Ego and shadow

What the organization says it is, and what it will not say about itself.

The ego register

The organization as it knows and presents itself.

Mission, stated values, official strategy, the brand narrative, the founding story, owned competences.

Built from stated material: policy, handbooks, decks, leadership messaging.

The shadow register

The organization as it does not yet know itself.

Systematically undiscussable topics, gaps between values and resource allocation, units scapegoated, failures retold as bad luck.

Built from indirect evidence, and from at least two independent sources.

- 1 Evidence or nothing** — No trait enters either register without pointing at stored percepts.
- 2 Confidence decays** — Unless refreshed. Contradicting evidence decrements. Below the floor, the trait is retracted.
- 3 The shadow is offered, never imposed** — It shapes the mind's questions, not its assertions — until the organization acknowledges the pattern itself.

Gotham — the open world

Liberty City. It runs whether or not the mind acts.

	Eidos	Gotham	Expanse
Time	turn-based	continuous	continuous
Scale	64×64 grid	a city	a planet
Others	none	thousands	everyone
Physics	none	simulated	real
Cost of error	an action	a restart	a robot

Gotham is the bridge between the grid and the body. Real-world spatial scale, real-world social density, real-world consequence structure — at simulation cost and simulation safety. A mind that can hold a city in a spatial ladder and navigate it by landmark is a mind ready for a warehouse.

Unity Mind

One mind. Many bodies. One memory.



Not a fleet of agents coordinating. One subject with several bodies.

The robot that learns a shelf layout and the organization model that knows why the shelf matters are the same mind, and neither has to tell the other — because a monad is a monad wherever it was minted.

What an enterprise does not have

Data systems, dashboards and models — but no subject.

Learns

the organization's REVEALED preferences — from budgets, calendars, promotions, and what actually ships

Models

ego and shadow, so guidance addresses how the organization behaves rather than how its mission statement says it does

Guides

toward opportunities scored against actual capability, and away from threats it is inflicting on itself

Journals

facilitated retrospectives, and prompts about what was avoided

Remembers

across years — durable knowledge base, and every inference carries its evidence

And because the Piagetian Modeler is world-agnostic, the same mind extends into the warehouse, the market, and the careers of the people in it.

Inspectable by construction

Every conclusion is a row, and the row names its evidence.

- | | | |
|---|--------------------|--|
| 1 | the percept | what arrived, from which Psyche, on which channel, at which moment |
| 2 | the match | which case it matched, and at what score, against which bar |
| 3 | the scheme | what was minted or recalled, and in which reality |
| 4 | the belief | the certainty, the finding, and whose viewpoint it is |
| 5 | the attempt | what was tried, with which parameters, under which token |
| 6 | the result | what came back, correlated by trial number |

The Curator records what the OBSERVER did, separately. Without that, an archive cannot distinguish an attempt the mind decided from one a person typed — and evidence that cannot be told apart from its operator's interference proves nothing.

Written in Premise

A language whose primitives are the things minds are made of.

```
(rule Recognise-Skill
  domain Deliberation
  salience 9600
  with [Competency :Name deliberating :Enabled yes]
        [Agenda ^ as ?agendum :Telon as ?telon]
        [Activation :M = ?telon
              :Reality = Desired :Pulses > 0]
  do
    (var ?fit (skill-serving ?telon))
    (unless (null-p ?fit)
      (enable :reifiers {?fit}
              :moment (moment) :reality Desired)))
```

relations what the mind holds

schemes premises and conclusions

agents mechanisms that run on a cycle

rules mechanisms that fire on a match

activations what is present, and in which reality

moments, beats wall time, and narrative time

Not a general-purpose language with a knowledge library bolted on.

github.com/subthought/premise/releases

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Latest Interview

youtube.com/watch?v=MVEz9QSOZjM

Building Sentient Beings

zenodo.org/records/15522356

Building Minds with Patterns

a.co/d/eqYOLBX

Coding Artificial Minds — hardcover

bit.ly/4gZs6bn

Coding Artificial Minds — paperback

bit.ly/3UZswXF

Premise Language

github.com/subthought/premise/releases

Cognitive Architectures

youtube.com/@CognitiveArchitectures